

VERDANT

An AI Sustainability Platform for Individuals and Communities.

Smarter choices. Healthier communities. A greener future.



SDG 11:
Sustainable
Cities



SDG 12:
Responsible
Consumption



SDG 13:
Climate Action

The Gap Between Awareness and Action

The Human Challenge

- People want to make sustainable choices but lack access to experts.
- Existing advice is generic, unactionable, and overwhelming.

The Tech Challenge

- Generative AI produces fluent, confident answers.
- When asked to calculate footprints, AI hallucinates completely unsupported environmental numbers.

Core Question: How can AI turn sustainability concerns into trustworthy, practical decisions without inventing the math?

Meet Verdant

Combining grounded AI guidance with deterministic environmental analysis.

Ask Verdant



User: What are some eco-friendly cleaning product alternatives for kitchen surfaces?



Verdant AI: Based on our analysis, consider these plant-based options:

1. Vinegar and water solution (Source: EPA Sustainable Management).
2. Baking soda paste (Source: Environmental Working Group).
3. Certified biodegradable multi-surface cleaners (Source: Green Seal).

[EPA] [EWG] [Green Seal]

[Find Local Suppliers](#)[Compare Costs](#)[Learn More](#)

Natural language questions → Grounded answers & sources.

Impact Analyzer



Everyday lifestyle inputs → Estimated impact & practical comparisons

Ask → Understand → Measure → Compare → Act

The Central Design Principle

“Numbers are calculated. Language is generated. The two never swap places.”

Engine:

Deterministic
Impact Engine.

Mechanism:

Fixed emission factors
+ Pure arithmetic.

Output:

Calculated Environmental
Estimates.

The AI never touches the math.
The calculator never writes the advice.

Engine:

IBM Granite via
watsonx.ai.

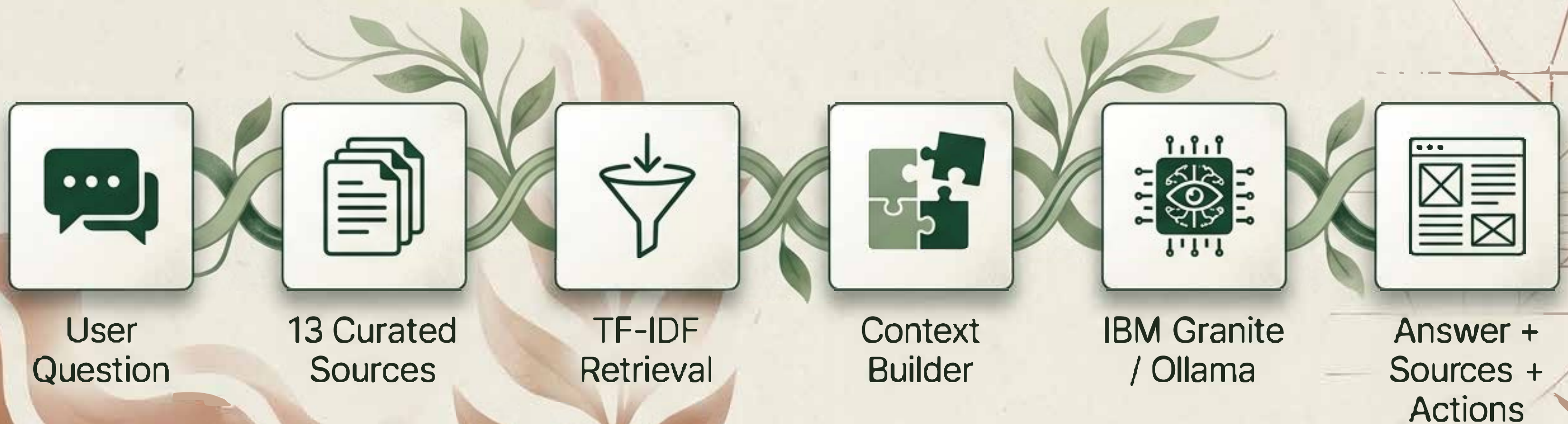
Mechanism:

Curated documents
+ Grounding.

Output:

Explanations &
Recommendations.

Ask Verdant: Grounded Context via RAG



Grounded Fallback System: If the 13-document knowledge base lacks the answer, the system is instructed to acknowledge the gap rather than improvise. It features a non-credentialed extractive fallback that runs entirely locally.

Impact Analyzer: See the True Cost of Choices

Inputs

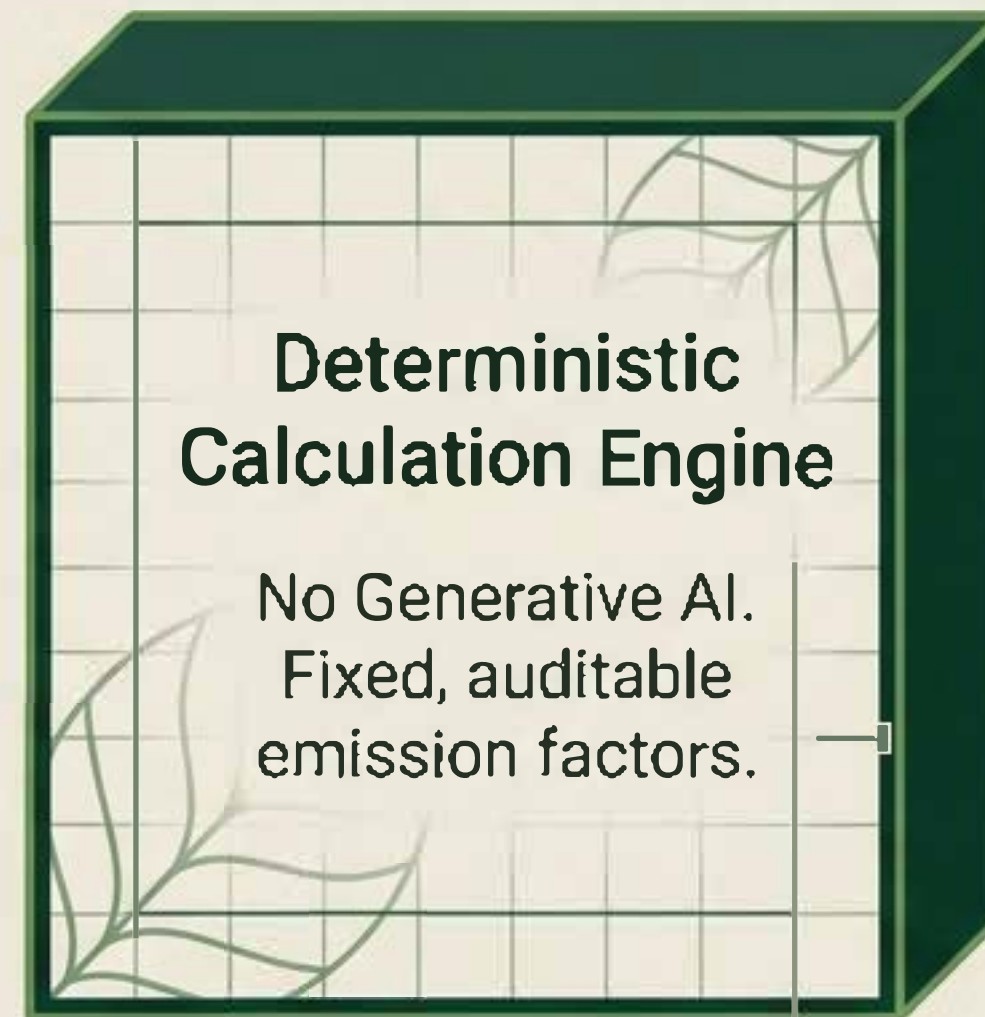
 Transportation

 Electricity

 Waste

 Habits

The Engine



Outputs

 Estimated annual impact
(planning-grade)

 Category breakdown

 Current vs. improved
scenarios

 Potential kg CO₂e savings



Responsible AI by Architecture, Not Just Policy

Transparency	Sources, publishers, and model providers are explicitly named alongside every UI response.
Privacy	No user accounts required; no server-side conversation storage or coordinate logging.
Grounded AI	Constrained system prompts prevent hallucination outside the 13 curated documents.
Human Agency	AI output is framed strictly as guidance and estimates, never absolute directives.
Honesty	The system explicitly self-identifies when relying on a fallback responder instead of Granite.

Full-Stack AI Sustainability Platform

Frontend

React + Vite + Tailwind (Biophilic UI, dependency-light, accessible)

API

FastAPI (Routing: /chat, /impact, /recommendations)

Generative AI

TF-IDF RAG + IBM Granite / Ollama

Biophilic, grounded responses via RAG & Granite;
fallback to Ollama for specific queries.

Deterministic Engine

Deterministic Impact Engine + Topic Catalogue

Fixed, auditable calculations for impact metrics,
ensuring accuracy and no generation.

Testing Anchor

Boundary Enforcement: 23 Pytest backend tests strictly verify the boundary
between calculated figures and generated text.

Verified Impact and Global Goals

Verification

23 backend tests
passed

Automated
calculation-integrity
checks

39 WCAG contrast
pairs validated

npm run verify
frontend smoke testing

Impact Goal

Transform vague
environmental
concerns into
understandable
information,
measurable
estimates, and
one practical
next action.

SDGs



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From Prototype to Community Platform

Future Roadmap

- 🌿 Dense semantic embeddings (Upgrading from TF-IDF)
- 🌿 Region-specific emission factors
- 🌿 Campus and community sustainability modes
- 🌿 Authentication & rate limiting for public deployment

The Verdant Philosophy

“Verdant doesn’t ask AI to invent the answer. It combines trusted information, verified calculations, and AI-powered explanation to help people make informed sustainable choices.”

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